Möbius-Ribbon Capacitance: Annulus, Conformal Modulus, and BIE Revisions of the α Estimate

Sequel to *The Möbius-Screw Electron*

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Abstract

The companion framed-unknot electron note [1] obtains $\alpha^{-1} \approx 137.036$ from an annulus / thin-ring capacitance and a self-stress match $(e/2)^2/(2C) = m_e c^2$. That note (§4.1) asks for a conformal-map and boundary-integral revision of C. This sequel implements three as-reported models at fixed spin-radius input $R = \hbar/(2m_e c)$: (i) the baseline annulus formula; (ii) a double-cover strip→annulus conformal-modulus image; (iii) constant-panel single-layer BIE collocation on a discretized Möbius ribbon about the $(2, 1)$ centerline. Aspect ratios are scanned; capacitances and α^{-1} are written to machine-readable results. All outputs are model claims under stated inputs. Comparison uses the CODATA-style reference $\alpha^{-1} \approx 137.035999084$; no uniqueness or derivation claim for α is made.

Contents

1 Scope and claim discipline

- **In scope:** numerical C and α^{-1} under three explicit capacitance models; aspect-ratio sensitivity; comparison to the CODATA-style reference value of α^{-1}.
- **Out of scope:** a proof that any model yields the measured fine-structure constant; QED radiative corrections; a first-principles prediction of a/R from the Standard Model.

Reference. Unless otherwise noted, numerical comparisons use

$$\alpha_{\mathrm{ref}}^{-1} \approx 137.035999084 \tag{1}$$

(CODATA-style). Runtime constants follow `scipy.constants` (float64), whose reciprocal fine-structure value is $\alpha^{-1} \approx 137.035999178$ in the recorded run.

2 Self-stress match (unchanged)

With charge split $e/2$ per double-cover sheet and rest-energy matching,

$$\frac{(e/2)^2}{2C} = m_e c^2. \tag{2}$$

Combined with $\alpha = e^2/(4\pi\epsilon_0 \hbar c)$ and $R = \hbar/(2m_e c)$,

$$\alpha^{-1} = \frac{\pi\epsilon_0 R}{C}. \tag{3}$$

Equation (3) converts any positive finite C into a model α^{-1}.

3 Three capacitance models

3.1 Annulus / thin-ring baseline

As in [1],

$$C_{\mathrm{ann}} = \frac{2\pi\epsilon_0 R}{\ln(8R/a) + 1}. \tag{4}$$

Under (3) this collapses to the closed form

$$\alpha_{\mathrm{ann}}^{-1} = \tfrac{1}{2}\big(\ln(8R/a) + 1\big). \tag{5}$$

Matching $\alpha_{\mathrm{ref}}^{-1}$ forces an extreme aspect ratio $a/R = 8\exp(1 - 2\alpha_{\mathrm{ref}}^{-1}) \sim 2 \times 10^{-118}$. That match is a tuning of the cutoff, not a geometric prediction of a.

3.2 Conformal double-cover strip → annulus

Take a rectangular strip of length $L = \text{cover} \cdot 2\pi R$ (default cover $= 2$) and width $W = 2a$. The exponential map sends the strip to an annulus with modulus

$$\rho = e^{2\pi W/L} = \exp\big(2a/(\text{cover}\, R)\big). \tag{6}$$

Geometric-mean radius $R_m = L/(2\pi) = \text{cover}\, R$ and effective half-width

$$a_{\mathrm{eff}} = R_m \sinh\big(a/(\text{cover}\, R)\big) \tag{7}$$

are inserted into the same logarithmic scheme as (4):

$$C_{\mathrm{conf}} = \frac{2\pi\epsilon_0 R_m}{\ln(8R_m/a_{\mathrm{eff}}) + 1}. \tag{8}$$

This is a conformal-modulus revision of the cutoff geometry, still within the thin-ring capacitance scheme.

3.3 BIE / collocation on a Möbius ribbon

Discretize the immersed ribbon

$$\mathbf{X}(u,v) = \mathbf{r}(u) + v\,\mathbf{n}(u), \qquad u \in [0, 2\pi), \quad v \in [-a, a], \tag{9}$$

about the $(2,1)$ centerline of [1], with $\mathbf{n}$ a torus-based Möbius half-twist frame (stable against Frenet flips). Constant-panel collocation of the single-layer potential at unit potential yields $C_{\mathrm{BIE}} = Q/V$. Lengths are nondimensionalized by R; the SI capacitance is $C = C\,\epsilon_0 R$. Mesh parameters in the recorded run: $n_u = 40$, $n_v = 5$ on the scan (demo point $n_u = 48$, $n_v = 6$). BIE is reported only on a moderate aspect window where the half-width is panel-resolved ($5 \times 10^{-3} \le a/R \le 0.25$).

4 Numerical results (as-implemented)

Code path: `rh_papers_may21/acs-framework/code/capacitance_ribbon/ribbon_capacitance.py`. Artifacts: `docs/capacitance_ribbon_results.json` and `docs/capacitance_ribbon_logs/`.

4.1 Worked demo at $a/R = 0.05$

Model	C [F]	α^{-1}	$C/(\epsilon_0 R)$
Annulus	1.768×10^{-24}	3.037587	1.034
Conformal (cover= 2)	3.174×10^{-24}	1.692054	1.857
BIE Möbius	1.441×10^{-23}	0.372688	8.430

At moderate aspect ratio the log-ring models sit near the thin-wire scale $C \sim \epsilon_0 R$, while the finite-width BIE ribbon sits near the disk scale $C \sim 8\epsilon_0 R$. Correspondingly, α^{-1} from (3) is $\mathcal{O}(1)$, not $\mathcal{O}(137)$.

4.2 CODATA-matching aspect ratios (analytic models)

Model	a/R at $\alpha^{-1}_{\mathrm{ref}}$	Note
Annulus	$\approx 2.039 \times 10^{-118}$	closed form
Conformal	$\approx 3.824 \times 10^{-237}$	root find; cover= 2
BIE	—	not resolved at that aspect

4.3 Ordering on the scanned grid

On the recorded a/R grid (including the BIE window),

$$\alpha^{-1}_{\mathrm{ann}} > \alpha^{-1}_{\mathrm{conf}} > \alpha^{-1}_{\mathrm{BIE}} \tag{10}$$

at shared aspect ratio: the conformal double-cover image increases C relative to the single-radius annulus scheme, and the immersed ribbon BIE increases C further. None of the moderate-aspect samples approach $\alpha^{-1}_{\mathrm{ref}}$; the annulus/conformal approach to the reference value requires the extreme cutoffs above.

Remark 1 (Reading of [1]'s 137.036)**.** *The companion value $\alpha^{-1} \approx 137.036$ is recovered by the annulus closed form at a tuned a/R. Under the conformal and BIE revisions implemented here, the same moderate geometric aspect does* not *reproduce that figure. This is the geometric-fidelity caveat of [1], §4.1, made quantitative.*

5 Summary

Object	Status
C_{ann} (4)	baseline; CODATA match at extreme a/R
C_{conf}	conformal-modulus revision; still log-ring scheme
C_{BIE}	immersed Möbius collocation; moderate-aspect only
α^{-1} via (3)	model output; not a uniqueness claim

Acknowledgments

This sequel answers the revision path stated in [1], §4.1 (geometric fidelity of the capacitance model).

References

[1] Flag Condensate Collaboration, *The Möbius-Screw Electron: Framed Unknot Geometry for $g = 2$ and a Capacitance Model of α*, preprint (July 2026), `mobius_screw_electron.tex` / `papers/notes/Mobius_Screw_Electron.tex`.